

Rajagopal Somaskandan

+91 7200288242 | somaskandan931@gmail.com | [linkedin.com/in/somaskandan-rajagopal](https://www.linkedin.com/in/somaskandan-rajagopal) | rajagopal-somaskandan.netlify.app

SUMMARY

Entry-level Machine Learning Engineer with hands-on experience building end-to-end AI systems using FastAPI, PyTorch, and modern ML frameworks. Built production-oriented projects in fraud detection, recommendation systems, computer vision, and privacy-preserving crowd analytics. Strong interest in backend ML systems, explainable AI, and scalable deployment.

EDUCATION

Sathyabama Institute of Science and Technology <i>M.Sc. Computer Science</i>	Expected 2027
Sathyabama Institute of Science and Technology <i>B.Sc. Computer Science (AI Specialization)</i>	2025

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL
Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Explainable AI
Deep Learning: PyTorch, Neural Networks, CNNs, RNNs, LSTMs, GRUs
NLP & Retrieval: Transformers, Embeddings, RAG, BERT, Sentence-Transformers, FAISS, ChromaDB
Backend & APIs: FastAPI, REST APIs, Streamlit
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB
Tools & Platforms: Git, GitHub, Docker, React

EXPERIENCE

Production AI Developer (Intern) <i>InnoXR Labs Pvt. Ltd.</i>	Mar 2026 – Jun 2026 <i>Chennai, India</i>
- Built and deployed ML-powered FastAPI microservices for real-time inference in AR and immersive applications. - Designed API-first ML systems integrating model inference with scalable backend endpoints. - Delivered end-to-end features from model development to deployment in a startup environment.	
Data Analyst Intern <i>India Advocacy</i>	Jul 2024 – Oct 2024 <i>Chennai, India</i>
Built Java and Selenium pipelines to scrape and process 1M+ records, transforming raw data into 15–20k structured datasets through multi-stage validation.	
Security Trooper (National Service) <i>Singapore Armed Forces</i>	Jan 2020 – Jan 2022 <i>Singapore</i>
- Executed security operations in regulated environments, ensuring strict protocol compliance and operational readiness. - Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.	
Sales Associate (Part-Time) <i>Bata</i>	Feb 2022 – Aug 2022 <i>Singapore</i>
- Handled high-volume customer interactions in a fast-paced retail environment, ensuring efficient service delivery. - Managed billing, inventory coordination, and issue resolution to support smooth store operations.	

PROJECTS

FinGuard Pro: Real-Time Fraud Detection & Compliance System	IEEE ICAaic 2025
- Built a fraud detection system using XGBoost, achieving 94% recall and ROC-AUC 0.995 on highly imbalanced financial data. - Integrated SHAP-based explainability to generate audit-ready insights for AML and compliance workflows. - Deployed backend services with FastAPI and a Streamlit dashboard for real-time monitoring and alerts. - Automated report generation in PDF and ZIP formats to reduce manual investigation effort.	

CareerGenie: AI Career Platform	Hackathon Winner, Origin 2026
- Developed an AI-powered career platform using RAG with MiniLM embeddings and ChromaDB for semantic job matching. - Implemented personalized recommendations using learning-to-rank (BPR) based on user feedback signals. - Built a multi-agent pipeline with LLaMA to improve response quality and reduce hallucinations. - Designed a full-stack system with a FastAPI backend and React frontend.	
MonoSplat: Single-Video 3D Reconstruction Pipeline	Internship Project, 2026
- Engineered a 3D reconstruction pipeline using COLMAP and Gaussian Splatting in PyTorch. - Designed a hybrid CPU-GPU pipeline, optimizing preprocessing with FFmpeg for cloud execution. - Built an async FastAPI backend with a Three.js viewer for interactive 3D visualization. - Focused on performance and usability, enabling smooth rendering and faster end-to-end processing.	
S.A.F.E: Privacy-Preserving Crowd Anomaly Detection	Design Thinking Project, 2026
- Developed a privacy-first crowd anomaly detection system using Farneback optical flow and zone-based aggregation to analyze motion without tracking individuals. - Evaluated unsupervised models including Isolation Forest, One-Class SVM, Z-Score, Moving Average Detector (MAD), and LSTM Autoencoder; MAD achieved strong cross-domain performance with ROC-AUC 0.8391 on the Mall dataset. - Reduced false positives by 40–60% using camera shake detection based on global motion thresholds. - Built a FastAPI backend and Streamlit dashboard that provide explainable, real-time alerts for safety operators.	
SourceUp: Explainable Supplier Recommendation System	Final Year Capstone, 2026
- Implemented semantic supplier search using Sentence-BERT and FAISS for natural language queries. - Designed a ranking model using Scikit-learn with cost, delivery time, and reliability constraints. - Added an explainability layer to provide feature-level insights for supplier selection and risk assessment.	

PUBLICATIONS & CONFERENCES

FinGuard Pro: Explainable AI for Financial Fraud Detection <i>ICAAIC 2025</i>	ISBN: 979-8-3315-6587-9
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ACHIEVEMENTS & HACKATHONS

Industry Innovation Hackathon <i>Sathyabama Institute of Science and Technology – National Level</i>	Mar 2026
Won Best Industry Innovation Award at Industry Innovation '26.	
Origin 24-Hour Hackathon <i>SIMATS Engineering, Chennai</i>	Apr 2026
Placed 4th at Origin 24-Hour Hackathon.	